

Categories, Proofs, and Processes: Assignment 5

Wira Azmoon Ahmad

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Question 1

(a)

The set-indexed product of a set of objects $\{A_i\}_{i \in I}$ is the Cartesian product

$$\prod_{i \in I} A_i = \left\{ g : I \rightarrow \bigcup_{i \in I} A_i \mid \forall i \in I : g(i) \in A_i \right\}$$

with its arrows $\pi_j(g) = g(j)$ for all $j \in I$.

Let C be an object with a set of arrows $f_i : C \rightarrow A_i$. Then the unique arrow $\langle f_i \rangle_{i \in I}$ is defined where for all $c \in C$,

$$\langle f_i \rangle_{i \in I}(c) := i \mapsto f_i(c)$$

For each $j \in I$, and for all $c \in C$,

$$\begin{aligned} \pi_j \circ \langle f_i \rangle_{i \in I}(c) &= \pi_j(i \mapsto f_i(c)) \\ &= (i \mapsto f_i(c))(j) \\ &= f_j(c) \end{aligned} \tag{1}$$

so $\pi_j \circ \langle f_i \rangle_{i \in I} = f_j$.

Suppose there is another arrow $h : C \rightarrow \prod_{i \in I} A_i$ such that $\forall j \in I, \pi_j \circ h = f_j$. Then $\forall c \in C$,

$$\begin{aligned} \pi_j \circ h(c) &= \pi_j(h(c)) \\ &= h(c)(j) \\ &= f_j(c) \\ &\implies h(c) :: j \mapsto f_j(c) \end{aligned} \tag{2}$$

So $h = \langle f_i \rangle_{i \in I}$ and the arrow is unique.

(b)

When $|I| = 2$, we get binary products. The product of 2 objects A_i, A_j for $i, j \in I$ is the object $A_i \times A_j$ with arrows $A_i \xleftarrow{\pi_i} A_i \times A_j \xrightarrow{\pi_j} A_j$ such that for any object C and set of arrows $A_i \xleftarrow{\pi_i} C \xrightarrow{\pi_j} A_j$, there is a unique arrow $\langle f_i, f_j \rangle : C \rightarrow A_i \times A_j$ such that $\pi_i \circ \langle f_i, f_j \rangle = f_i$ and $\pi_j \circ \langle f_i, f_j \rangle = f_j$.

When $|I| = 1$, we get terminal objects. The product of 1 object A_i for $i \in I$ is the terminal object A_i with the identity 1_{A_i} such that for any object C , the arrow $f_i : C \rightarrow A_i$, is the unique arrow to A_i .

(c)

Let (L, γ) be the limit of $D : \mathcal{I} \rightarrow \mathcal{C}$. Then the arrows are defined as

$$\{\gamma_J : L \rightarrow D J\}_{J \in \text{Ob}(\mathcal{I})}$$

where the only arrows in $\mathcal{J}$ are identities 1_J so $D 1_J = 1_{D J}$. Since (L, γ) is the terminal object in $\mathbf{Cone}(D)$, for every other cone to D , (A, δ) , there is a unique arrow $g : A \rightarrow L$ such that $\forall I \in \text{Ob}(\mathcal{I}), \gamma_I \circ g = \delta_I$.

By using the functor $S : \mathbf{Set} \rightarrow \mathbf{Cat}$ from Assignment 3 Exercise 1, let $S I = \mathcal{I}$, so there's an isomorphism $\phi : I \cong \text{Ob}(\mathcal{I})$. Then set $\forall j \in I$,

- $D \phi(j) := A_j$
- $L := \prod_{j \in I} A_j$
- $\gamma_{\phi(j)} := \pi_j$
- $A := C$
- $\delta_{\phi(j)} := f_j$
- $g := \langle f_j \rangle_{j \in I}$

so the limit of the diagram matches the definition of the set-indexed products.

(d)

Suppose $\mathcal{D}$ has all set-indexed products.

Let I be the set of indices, and $\{F_i : \mathcal{C} \rightarrow \mathcal{D}\}_{i \in I}$ be a set of functors. First, we construct the product functor $\prod_{i \in I} F_i : \mathcal{C} \rightarrow \mathcal{D}$ and prove its functoriality.

$$\prod_{i \in I} F_i(C) := \prod_{i \in I} (F_i C)$$

$$\prod_{i \in I} F_i(f : C \rightarrow C') := \prod_{i \in I} (f_i : F_i C \rightarrow F_i C') := \langle f_i \circ \pi_i \rangle_{i \in I}$$

where $\pi_i : \prod_{i \in I} F_i \rightarrow F_i C$, which exists in $\mathcal{D}$.

Let $C \xrightarrow{f} C' \xrightarrow{g} C''$ be arrows.

$$\begin{aligned}
\prod_{i \in I} F_i(g \circ f) &= \langle (g \circ f)_i \circ \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} (g \circ f)_i \circ \langle \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} g_i \circ \prod_{i \in I} f_i \circ \langle \pi_i \rangle_{i \in I} \\
&= \langle g_i \circ \pi_i \rangle_{i \in I} \circ \prod_{i \in I} f_i \circ \langle \pi_i \rangle_{i \in I} \\
&= \langle g_i \circ \pi_i \rangle_{i \in I} \circ \langle f_i \circ \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} F_i(g) \circ \prod_{i \in I} F_i(f)
\end{aligned} \tag{3}$$

$$\begin{aligned}
\prod_{i \in I} F_i(1_C) &= \langle 1_{F_i C} \circ \pi_i \rangle_{i \in I} \\
&= \langle \pi_i \rangle_{i \in I} \\
&= \prod_{i \in I} 1_{F_i C} \\
&= 1_{\prod_{i \in I} (F_i C)} \\
&= 1_{\prod_{i \in I} F_i(C)}
\end{aligned} \tag{4}$$

So the product functor satisfies functoriality.

Next, we show the naturality of the functor projections. Let $f : C \rightarrow C'$ be an arrow in $\mathcal{C}$, and $f_j : F_j C \rightarrow F_j C'$ be an arrow in $\mathcal{D}$. For an arbitrary natural transformation $t_j : \prod_i F_i \rightarrow F_j$, define $t_{j,C} = \pi_j : \prod_i F_i C \rightarrow F_j C$, and $t_{j,C'} = \pi'_j : \prod_i F_i C' \rightarrow F_j C'$, which both exist as projections in $\mathcal{D}$. Then by the property of the product, we have

$$f_j \circ \pi_j = \pi'_j \circ \langle f_i \circ \pi_i \rangle_i$$

So naturality is satisfied.

Let again $f : C \rightarrow C'$ be an arrow in $\mathcal{C}$. Suppose we have some functor G and a set of natural transformations $s_i : G \rightarrow F_i$. Define the natural transformation $\langle s_i \rangle_i : G \rightarrow \prod_i F_i$ such that

$$\langle s_i \rangle_{i,C} = \langle s_{i,C} \rangle_i : G C \rightarrow \prod_i F_i C$$

By naturality of s_i , we have

$$f_i \circ s_{i,C} = s_{i,C'} \circ G f$$

From the property of the product, we have that

$$\pi_j \circ \langle s_i \rangle_{i,C} = s_{j,C}, \quad \pi'_j \circ \langle s_i \rangle_{i,C'} = s_{j,C'}$$

Then these imply that

$$\begin{aligned}
f_j \circ \pi_j \circ \langle s_i \rangle_{i,C} &= \pi'_j \circ \langle s_i \rangle_{i,C'} \circ G f \\
\pi'_j \circ \langle f_i \circ \pi_i \rangle_i \circ \langle s_i \rangle_{i,C} &= \pi'_j \circ \langle s_i \rangle_{i,C'} \circ G f
\end{aligned}$$

Since this is true for all $i \in I$, then by the property of the product,

$$\langle f_i \circ \pi_i \rangle_i \circ \langle s_i \rangle_{i,C} = \langle s_i \rangle_{i,C'} \circ G f$$

So the naturality of $\langle s_i \rangle_i$ is satisfied. Since $\forall C$ in $\mathcal{C}$, and $\forall i \in I$, we have some π_i that

$$\pi_j \circ \langle s_i \rangle_{i,C} = s_{j,C}$$

where $t_j = \pi_j$, then

$$\forall j \in I : t_j \circ \langle s_i \rangle_i = s_j$$

Now, we show the uniqueness of the natural transformation. Suppose there is some other natural transformation $u : G \rightarrow \prod_i F_i$ such that $\forall j \in I, t_j \circ u = s_j$. Then for all C in $\mathcal{C}$, $\forall j \in I, \pi_j \circ u_C = s_{j,C}$ for some π_j . Then there is a set of mappings

$$\pi_j \circ u_C = s_{j,C} : G C \rightarrow F_j C$$

By property of the product, there is a unique mapping

$$\langle \pi_i \circ u_C \rangle_i = \langle s_{i,C} \rangle_i : G C \rightarrow \prod_i F_i C$$

such that $\forall j \in I, \pi_j \circ \langle \pi_i \circ u_C \rangle_i = s_{j,C}$. Then for all C in $\mathcal{C}$,

$$\begin{aligned} \langle \pi_i \circ u_C \rangle_i &= \langle s_{i,C} \rangle_i \\ \langle \pi_i \rangle_i \circ u_C &= \langle s_i \rangle_{i,C} \\ 1_{\prod_i F_i C} \circ u_C &= \langle s_i \rangle_{i,C} \\ u_C &= \langle s_i \rangle_{i,C} \\ u &= \langle s_i \rangle_i \end{aligned} \tag{5}$$

Thus, $\langle s_i \rangle_i$ is the unique arrow.

(e)

Question 2

(a)

(b)

(c)

(d)

(e)

(f)